

07b · The same campaign question, on real data — Google’s geo experiment (companion)

July 13, 2026

The business decision. A real advertiser spent **\$50,000** on a real ad campaign, in **50 US geos, for four weeks** (16 Feb – 15 Mar 2015), and left 50 geos dark. This is the worked example published with Google’s geo-experiment methodology (Kerman, Wang & Vaver 2017, the `matched_markets` library) — daily **sales and ad cost per geo**, exactly the panel your own company’s data warehouse holds. The question is the one notebook 07 trained us for: **how much incremental sales did the spend cause, and did it pay for itself?**

What changes when the data is real

Notebook 07 planted a +12% lift in a simulator and graded synthetic control against it. Here nobody tells us the truth — that is the whole epistemic situation of applied causal inference. Three things replace the planted truth:

1. **A randomized benchmark.** The 100 geos were split into treatment and control **at random** (in matched pairs, as we’ll verify from the data itself). So a design-based estimate — the matched-pairs difference-in-differences — is *unbiased by construction*. (Recall 07’s difference-in-differences: the *change* in the treated group, post minus pre, minus the same change in the control group; randomizing which geos are treated makes that difference unbiased for the campaign effect, because the two groups can then differ only by the coin flip.) It is noisy, but it is the closest real data ever gets to a known answer, and we grade the synthetic control against it.
2. **Falsification tests** that need no truth: placebo campaigns on untreated geos (Depth B), a fake launch date, and a check the simulator could never give us for free — the campaign *ends*, so the estimated effect must **switch off when the money does**.
3. **Real messiness:** missing geo-days, heavy-tailed market sizes (the largest geo sells 300× the smallest), and a weekly cycle you can see with the naked eye. Each one bites somewhere below, and one of them (market size) breaks a naive estimator in Depth A.

How this notebook is organised

The same **7-step contract** as every notebook — with step 2’s “simulate a ground truth” honestly replaced by “load real data, and say what can still be validated” — plus the anchor depths:

- **Depth A** — the naive estimators on a *single* real market (where they fail for real), and a convex-hull failure the pre-fit gate catches
- **Depth B** — placebo inference: pseudo-cell permutation, the scale-free RMSE-ratio, placebo-in-time

- **Depth C** — the dollar decision and its sensitivity to the launch date, the donor pool, and the margin

FAST=False SC sampling={'draws': 1500, 'tune': 1500, 'chains': 4} pseudo-cells=200

2 · Load the real data (what replaces “simulate a ground truth”)

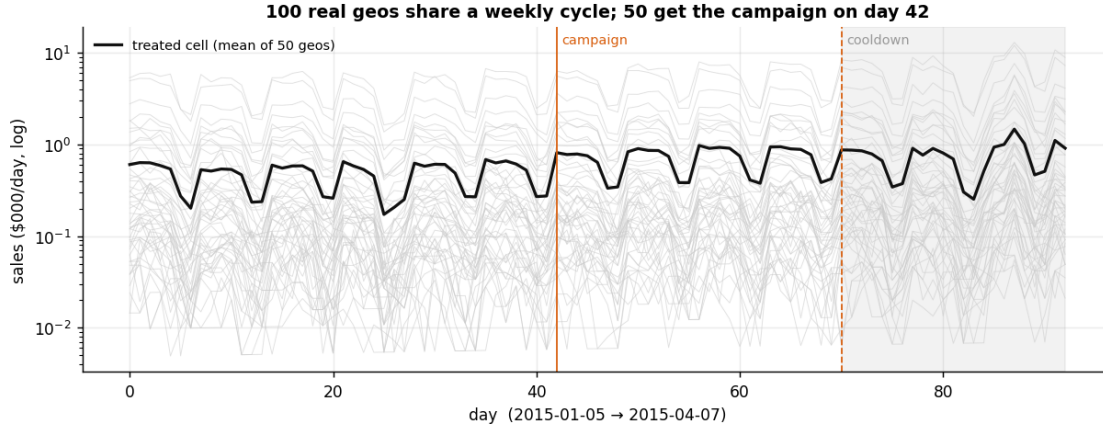
`cmp.data.load_google_geo()` fetches the two public CSVs (Apache-2.0, pinned commit; ~230 KB — cached under `~/.cache/cmp` after the first call) and returns the long panel: `date`, `geo`, `sales`, `cost`, `group`. The **data card**, verified in code below:

fact	value
units	100 US geos, numbered by size (geo 1 = largest — visible in the data)
period design	93 days, 2015-01-05 → 2015-04-07, daily randomized in matched pairs : consecutive size-ranked geos split 1 treated / 1 control
treatment	ad campaign in the 50 group-2 geos, 16 Feb – 15 Mar (28 days), \$50,000 total spend
cooldown	23 days <i>after</i> the spend stops — a free falsification window
blemishes	75 of 9,300 geo-days missing at source (small geos, mostly pre-period) — we interpolate and disclose

Two working conventions. **(a)** We analyse sales in **\$000s** — same numerical scale the fitters were built for. **(b)** The treated unit is the **test cell**: the *mean* daily sales per treated geo. The mean (not the sum) keeps the treated series inside the donors’ range — a sum of 50 geos could never be a convex combination of single geos (the synthetic counterfactual is only ever a weighted average of the donor geos, so it can never rise above the biggest donor or fall below the smallest — the treated series must sit *inside* the donors’ range, their ‘convex hull’), and Depth A shows what happens when a series escapes the donor hull.

Eyeball the panel below before any model: every geo rides the same weekly cycle (the shared factors of 07, no longer a modelling assumption but a visible fact), and the treated cell visibly jumps *inside the campaign window only*.

100 geos (50 treated / 50 control) * 93 days * campaign days 42-69 (28 days) *
cooldown 23 days
interpolated geo-days: 75/9300 (0.8%) - small geos, mostly pre-period
ad spend: \$0 before launch (no anticipation), \$50,000 inside the window, \$0
after - the design is clean
geo sizes are heavy-tailed: largest \$6,728/day vs smallest \$18/day (~369x)



3 · Identify — the estimand, and what randomization buys us

The estimand. Total incremental sales caused by the campaign in the treated geos over the 28-day window:

$$\tau = \sum_{t=T_0}^{T_1} \left(Y_{\text{cell},t}(1) - Y_{\text{cell},t}(0) \right) \times 50,$$

where $Y_{\text{cell},t}$ is mean sales per treated geo and the $\times 50$ restores cell totals. Recall 07’s potential-outcome notation: $Y_{\text{cell},t}(1)$ is sales *with* the campaign (what we observe in the treated geos) and $Y_{\text{cell},t}(0)$ is what those same geos *would* have sold with no campaign — the never-observed counterfactual the synthetic control must reconstruct. The estimator is exactly 07’s: simplex-weighted donors fit to the pre-period fill in the missing $Y_{\text{cell},t}(0)$ (see 07 §3 for the full math and the convex-hull argument). The dollar decision then divides by spend: **iROAS** = $\tau / \$50\text{k}$, incremental revenue per ad dollar.

The assumptions, upgraded by the design. In 07 most rows of this table were unverifiable promises; here randomization converts the worst one into a guarantee:

assumption	status on this real dataset
convex hull / no extrapolation	<i>checked:</i> the treated cell (a mean of 50 geos) sits mid-range of the donors; Depth A shows geo 1 violating it and the gate catching it
good pre-fit	<i>checked:</i> 42 pre-days, PASS/FAIL gate in Step 5
no anticipation	<i>checked from data:</i> \$0 of spend before day 42
stable co-movement	<i>supported:</i> shared weekly cycle visible; placebo-in-time (Depth B) probes it
no spillover between geos	<i>assumed:</i> geos are distinct media markets, but adjacency isn’t in the data — the honest caveat that survives even randomization

assumption	status on this real dataset
no concurrent treated-only shock	guaranteed in expectation by randomization — a shock would have to coincide with a random coin flip across 50 pairs; in 07 this row was “institutional knowledge only”

Why bother with synthetic control if the design is randomized? Because the design-based estimate, while unbiased, is *noisy* — geo sizes are so heavy-tailed that a plain treatment-vs-control comparison has a wide interval (Step 5 quantifies this). Synthetic control re-weights donors to *match* the treated cell and cuts that variance. And the moment you leave experiments — one treated market, chosen not by coin flip — SC is the only member of this family still standing. That is the real lecture point of this notebook: **the experiment validates the method we’ll use when there is no experiment.**

Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst does here **without** it. On this dataset the answer is not one estimator but **two**, and the difference between them is the whole lesson of a real-data companion.

The referee: the matched-pairs difference-in-differences. The 100 geos were split into treatment and control **by coin flip, within size-matched pairs**. That design licenses an estimator that needs **no model whatsoever** — no donor weights, no counterfactual to reconstruct, no assumption we cannot check. Take each pair’s treated geo, take its change in mean daily sales from the pre-window to the campaign window, subtract the same change for its size-matched control partner, and average over the 50 pairs:

$$d_p = (\bar{Y}_{p,\text{post}}^{\text{tr}} - \bar{Y}_{p,\text{pre}}^{\text{tr}}) - (\bar{Y}_{p,\text{post}}^{\text{ct}} - \bar{Y}_{p,\text{pre}}^{\text{ct}}), \quad \hat{\tau}_{\text{DiD}} = \underbrace{50 \times 28}_{\text{geos} \times \text{days}} \cdot \frac{1}{50} \sum_{p=1}^{50} d_p, \quad \widehat{\text{SE}}(\hat{\tau}_{\text{DiD}}) = 50 \times 28 \cdot \frac{s_d}{\sqrt{50}},$$

with s_d the sample standard deviation of the 50 pair differences. **The covariance choice, and why.** The SE is built from the *scatter of the 50 pair differences* — that is the design’s own variability, the noise the coin flip actually injects, and it is the right unit of clustering because randomization happened **at the geo level, within pairs**. A geo-day-level iid SE would treat 2,800 correlated geo-days as independent draws and report an interval several times too narrow. And *pairing itself is variance reduction*: each d_p compares a geo to its size-matched neighbour, so the enormous size factor (the largest geo sells $\sim 370\times$ the smallest) differences out **within** every pair instead of inflating s_d . The cell below prints what that buys, by re-running the identical contrast while *ignoring* the pairs.

This estimator is **unbiased by construction**. It is the closest thing to truth that real data ever hands you, and it is the thing that replaces notebook 07’s planted lift:

On real data there is no planted truth, so the grade is the design. Every model-based estimate below — classical synthetic control, and the Bayesian posterior

of §4 — must land inside the randomized anchor’s interval. The anchor is the referee; the models are the contestants.

The contestant: classical synthetic control. The same estimator 07 introduced, and the one we will still have when there is *no* experiment — pick the simplex weights over the 50 control geos that best reproduce the treated cell’s pre-launch path, by constrained least squares (SLSQP), then read the post-launch gap:

$$\hat{w} = \arg \min_{w \in \Delta^{J-1}} \sum_{t < T_0} \left(Y_{\text{cell},t} - \sum_j w_j Y_{jt} \right)^2, \quad \hat{\tau} = 50 \cdot \sum_{T_0 \leq t \leq T_1} \left(Y_{\text{cell},t} - \sum_j \hat{w}_j Y_{jt} \right).$$

Deterministic, milliseconds, no likelihood, no prior, no chain — and it is the *same objective* §4 will re-house inside a Dirichlet prior. (Full derivation and the convex-hull argument: [07 Step 0](#).)

Where does the SC arm’s interval come from? Not from a standard error — there is no likelihood to read one off, and only **one** treated cell. Abadie’s answer is **permutation inference**, adapted here to the design: the placebo unit is not a single geo but a **pseudo-cell** — 25 randomly chosen *control* geos bundled together and fitted a synthetic control from the other 25. None of them saw an ad dollar, so their post-window gaps map out the distribution the estimator produces from **noise alone**. The treated cell’s **rank** in that distribution is the p-value; subtracting the null’s 5th/95th percentiles from our estimate **inverts** it into a 90% *randomization interval* — design-based, assuming no shape for the errors, and the honest scale §5x will hold the posterior’s band up against. Two of Abadie’s refinements are used: the scale-free **post/pre RMSE ratio** as the test statistic, and a **pre-fit filter** that discards pseudo-cells the synthetic cannot track *before* launch (here: pre-RMSE more than 5× the real cell’s). Watch what the filter throws away — the print-out names it, and it is the very same pathology Depth A stages on purpose. Depth B opens the whole placebo box (spaghetti, RMSE-ratio histogram, placebo-in-time); Step 0 needs only its headline numbers.

THE REFEREE - matched-pairs DiD over 50 randomized pairs.

No model, no donor weights, no counterfactual: only the coin flip.
total incremental sales \$147k 90% CI [\$78k, \$216k] (SE \$42k)
covariance: matched-pairs randomization; SE from the 50 pair differences
iROAS 2.94 [1.56, 4.32] on \$50k of spend
what PAIRING bought: ignore the pairs and the same contrast gives the
same point (\$147k) with SE \$99k -> pairing cuts the SE 2.4x, by design
UNBIASED by construction - and still +/- \$69k wide.
Unbiasedness is NOT precision: that width is the variance the model cuts.

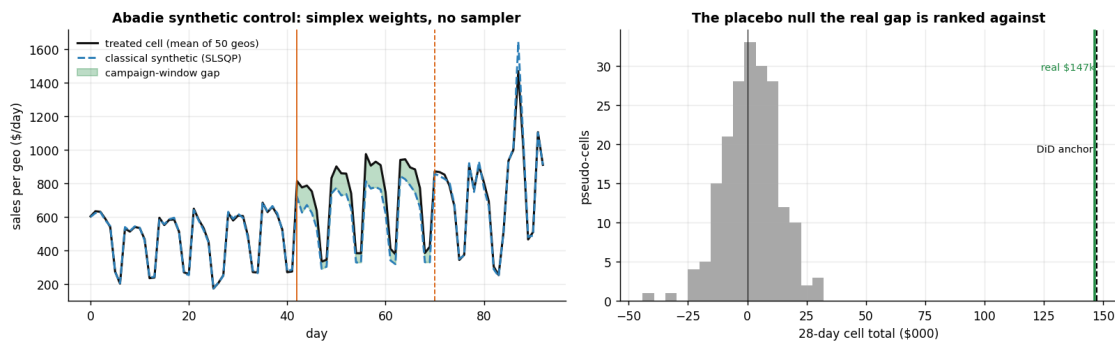
THE CONTESTANT - classical SC (SLSQP), fit on days 0-41 only

50 donors: 19 carry >1% of the weight; 16 effective donors (1/sum w^2)
pre-fit RMSE \$7/geo/day on a ~\$472 level (1.5%) - the gate the gap rests on
total incremental sales \$147k * iROAS 2.93 * \$+105/geo/day
GRADE vs the randomized referee: off by \$-1k (0.01 anchor SEs)
-> INSIDE the anchor's 90% CI

INFERENCE - no likelihood, so no standard error. Rank the real cell
against pseudo-cells that never saw an ad dollar, then
invert that null into an interval:

Abadie RMSE-ratio $p = 0.005$ (real cell 16.1 vs a placebo median of 1.8)
 raw-gap permutation $p = 0.010$ (1 of 200 pseudo-cells beat it)
 the pre-fit filter discards 7 of 200 pseudo-cells (pre-RMSE up to 42x
 the real cell's). The worst carries a |total| of \$481k - a bundle whose
 own synthetic cannot reach it: Depth A's convex-hull failure,
 arriving uninvited. Filtered, the raw-gap p becomes 0.005.
 90% randomization interval on the total (placebo null, inverted):
 [\$127k, \$164k] - width \$37k, vs the DiD referee's \$138k

What this 90% confidence interval does NOT say: that there is a 90% probability
 the true effect lies inside it. It is a property of the *procedure* - 90% of
 intervals built this way would cover the truth across repeated samples. This
 interval either contains the truth or it does not. To get a probability *about*
 the effect itself* - the thing a go/no-go rule like $P(\text{lift} > \text{cost}) \geq 0.9$
 actually needs - you need a posterior. That is what the Bayesian section adds.



Read-out — no priors, no sampler, and the question is already answered. Read the three blocks above in order. The **referee** says the campaign generated the total printed on the first line, an iROAS near 3 — and it says so with no model at all, only the coin flip. Note the two honest details the print-out volunteers: pairing cuts the standard error by the factor shown (a *design* choice worth more than any estimator choice), and the anchor is nonetheless **wide** — its 90% interval reaches nearly half the effect's size in each direction. Unbiasedness is not precision. That width is the entire commercial reason to fit a model on top of a perfectly good experiment.

The **contestant** — classical synthetic control, the estimator we will still own when nobody randomizes anything — tracks the treated cell before launch to within the printed pre-fit RMSE (~1.5% of the level), and then lands within a fraction of one anchor SE of the referee's point, **inside** its 90% interval. That is the exam, and it is the only exam real data can set: 07 could grade against a planted truth, and here the randomization *is* the truth we grade against. The method passes it.

The **inference** block is where the classical arm shows its real character. The permutation p-value is a **randomization-inference** statement: *if the campaign had done nothing, a post-window jump this extreme — relative to its own pre-fit — would have turned up in about this fraction of untreated bundles.* Watch what the pre-fit filter caught: the single pseudo-cell that beat the real cell on raw gap is one whose own synthetic cannot track it at all (pre-RMSE tens of times the real cell's). It is not evidence of a rival effect, it is a **broken fit** — the same convex-hull failure Depth A stages

deliberately with geo 1 — and Abadie’s filter, plus the scale-free RMSE ratio, is precisely the machinery that refuses to be fooled by it. Filter it and the raw-gap p falls to the floor of $1/(N+1)$.

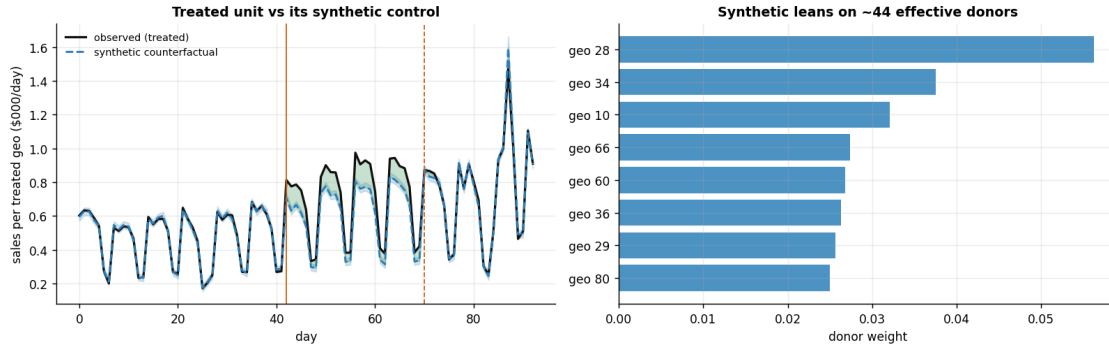
And now the boundary — the same one this cookbook states in every notebook. Both classical intervals above are *confidence* intervals, and the guardrail printed at the end of the cell applies to each of them: 90% of intervals built that way cover the truth **across repeated randomizations**; *this* one either covers it or it does not. The permutation p is stricter still — it ranks the observed data against a null and attaches **no probability to any hypothesis about the effect**. So neither object can be read as “there is a 90% chance the lift is between these numbers,” and — the part that actually bites — **neither can answer the question §6 has to sign off on**: *what is the probability this campaign was profitable at a 35% gross margin?* That is a probability **about the effect**, weighed against a cost the p-value has never met. It exists only in a posterior. §4 builds one; §5x then asks, with the design-based scale we just constructed, whether it is a *trustworthy* one.

4 · Estimate — Bayesian synthetic control on the test cell

Same model as 07 — and the same objective Step 0 just solved with SLSQP, now re-housed in a prior rather than an optimizer: **Dirichlet-simplex weights** over the 50 control geos, fit to the 42 pre-launch days, returning a posterior over the counterfactual path — and therefore over total lift and iROAS. Read three things off the fit, exactly as in 07: **which donors** carry weight (the Dirichlet spreads it — expect a high “effective number of donors” on a mean-of-50 target), the **pre-fit RMSE** (Step 5 turns it into a PASS/FAIL gate), and the **counterfactual band** (a posterior predictive, carrying the cell’s own daily noise, not just weight uncertainty).

And read the convergence line — including the divergences. The sampler already runs at `target_accept = 0.95`, yet NUTS still reports some divergent transitions (the exact count, r-hat, and ESS are printed below). The geometry is the suspect: 50 simplex weights pressed against a noise scale of ~2% of the level create funnel-like corners the integrator occasionally overshoots. We do *not* wave this away — divergences are the one warning that can invalidate a fit even when r-hat looks perfect. Two checks argue these are benign: the r-hat/ESS readout below says the chains agree on the same posterior (**r-hat** near 1.0 means the independent chains landed on the same distribution; **ESS** — effective sample size — is the number of *effectively independent* draws behind each estimate, and the several thousand (~3,300) printed below is healthy), and — decisive — Step 0 already solved the *same* objective with a divergence-free optimizer (SLSQP samples nothing, so it cannot diverge), and Step 5 prints how far the two totals sit apart (a small fraction of the posterior’s own 90% width). If that cross-check disagreed, the right move would be reparameterization, not narration.

```
SC sampling convergence: max r-hat 1.000 - min ESS 3268 - divergences 56
pre-fit RMSE $10/geo/day on a ~$472 level (~2%)
total incremental sales $149k 90% CI [$141k, $158k] on $50k of spend
revenue iROAS 2.98 90% CI [2.81, 3.15]
```



5 · Validate — against the randomized experiment

Four checks. The first is the one no simulator can fake, and Step 0 has already built it: an *unbiased, design-based* estimate of the same estimand, from the experiment’s own randomization.

1. **The matched-pairs DiD anchor** (Step 0). It is our stand-in for 07’s planted truth: **the synthetic control point must land inside its interval** — and the chart below draws the anchor straight across the gap path so you can see the daily lift agree with it. Note again how *wide* that interval is: unbiasedness isn’t precision, and cutting exactly that variance is what the model is here to do. (§5x then turns the comparison around and asks whether the model’s *narrowness* is honest.)
2. **Pre-fit gate** — same concrete rule as 07: pre-RMSE below a third of the daily lift the design must detect, printed PASS/FAIL.
3. **The off-switch falsification.** The spend stops on day 70; a real advertising effect should collapse toward zero in the 23-day cooldown (up to carryover). A gap that *persists* after the money stops would smell of a confound rather than a campaign. This replaces the simulator’s “we know the truth after week 40” with a physical prediction the data can refute.
4. **Leave-one-out donors** — drop each top-weighted donor, refit (fast SLSQP), watch stability.

What we cannot do on real data — and say so plainly, as 07’s multi-seed check taught us: we cannot refit on fresh copies of the world, so we cannot *measure* interval calibration here. But we are not reduced to quoting 07’s warning either: Step 0’s placebo distribution is the same estimator run under a known null, and **§5x prices the posterior’s band against it** — the closest thing to a calibration check that one real dataset permits. Trust the placebo permutation (Depth B) for significance, not the band width alone.

anchor (Step 0): matched-pairs DiD total \$147k +/- 42 (90% CI [78, 216]k; iROAS 2.94 [1.56, 4.32])

SC total \$149k (iROAS 2.98) - lands on the anchor's point with a ~8x tighter band - the variance reduction SC exists for. Is that narrowness honest? 5x prices it.

pre-fit gate: pre-RMSE \$10 vs |daily lift|/3 = \$36 -> PASS

off-switch: campaign-window gap \$+107/geo/day -> cooldown \$-3/geo/day

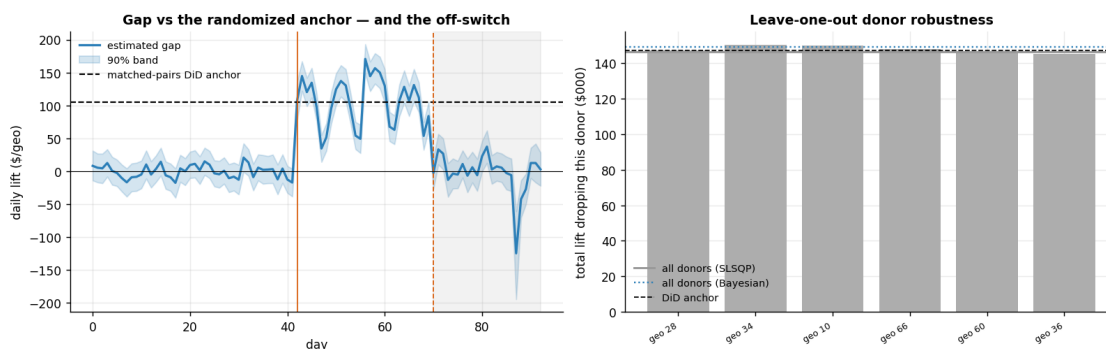
(collapses when the spend stops - and no long carryover to book as extra value)

leave-one-out: total ranges \$145-150k vs SLSQP baseline \$147k - no single donor drives it

divergence cross-check: 56 divergent transitions, but the divergence-free SLSQP refit of the same objective lands \$2.5k from the Bayesian total (15% of the 90%-CI width) -> benign (simplex funnel corners, not bias)

interpolation check (a): dropping the 10 gappy donors (near-uniform Dirichlet weight 19%) moves the daily lift \$+105 -> \$+105/geo/day (0.2% shift) -> interpolation is immaterial

interpolation check (b): dropping all 17 gappy geos on both sides gives \$+121/geo/day for the surviving 43-geo cell - pure composition (bigger geos, bigger per-geo lift), matched by the anchor on the same pairs (\$+121); relative lift 22.2% vs 22.3% - unchanged



Read-out. The synthetic control lands almost exactly on the randomized answer — **\$149k vs the anchor's \$147k** (iROAS 2.98 vs 2.94) — passing the only exam real data can set. And the comparison teaches why the model exists at all: the *unbiased* anchor carries a wide 90% interval (heavy-tailed geo sizes make even 50 randomized pairs noisy), while SC's band is several times tighter around the same point — a narrowness \$5x will hold up against the design's own scale before we spend it. The gap path shows a clean rectangle: nothing before day 42, a sustained ~\$105/geo/day lift during the four paid weeks, and a collapse to $\approx -\$3$ /geo/day within the cooldown — the effect **switches off when the money does**, which simultaneously validates the attribution and tells the media planner there is no adstock annuity to book. Leave-one-out totals move by ~\$3k on a \$147k estimate: no single market carries the result.

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0 produced two model-free numbers and an interval with no likelihood, prior or sampler; \$4 produced a posterior. All three target the **same estimand** — total incremental sales over the 28-day window — so lay them side by side on it. Then do the thing this notebook can do and 07 could only simulate: hold the posterior's interval up against a **design-based** scale for the very same quantity. We cannot resample the world here (no fresh panels, no planted truth), so we cannot *measure* coverage — but we do not have to guess either. The randomized anchor hands us an analytic SE, and the pseudo-cell placebos of Step 0 hand us the distribution *the same SC estimator* produces from noise alone. Those are the referees for **width**, exactly as the anchor is the referee for **location**. Nothing below is asserted; all of it is computed.

SAME estimand - total incremental sales over the 28 campaign days.
 No planted truth exists: the randomized DiD is the benchmark the two

model-based rows must match.

	est	iROAS	5%	95%	width
matched-pairs DiD * THE REFEREE (design)	147	2.94	78	216	138
classical SC (SLSQP) * placebo CI	147	2.93	127	164	37
Bayesian SC (Dirichlet) * posterior	149	2.98	141	158	17

LOCATION - they agree, and that agreement is the notebook's exam. Both model-based estimates sit INSIDE the referee's 90% CI: classical SC 0.01 anchor SEs from its point, the posterior mean 0.05.

The Bayesian layer moved the point estimate by \$2.5k (1.7%)
- it did not buy a better one.

WIDTH - they do not. Three rival scales for the sd of that same total:

posterior sd (iid likelihood, one dataset)	\$ 5.2k		
placebo sd - the SAME estimator, no ads	\$ 11.6k	(2.2x)	
deflated for the 25-vs-50-geo mismatch	\$ 8.2k	(1.6x)	<- fair floor
DiD anchor's SE (a DIFFERENT estimator)	\$ 41.9k	(8.1x)	

-> the posterior underprices the SC estimator's OWN sampling noise by ~1.6-2.2x. The anchor's 8x is NOT that verdict: an unweighted DiD is a genuinely noisier estimator, and cutting its variance is what SC is FOR. The fair referee re-runs OUR estimator under a known null - the placebos. 07, where fresh worlds were free, MEASURED this same failure: the posterior under-priced its own sd by ~2.8x, and the 90% band built from it covered

the

planted total 50% of the time, not 90%. Same direction, same cause: an iid likelihood on a gap whose errors are persistent -> the variance of a SUM is underpriced.

P(profitable at a 35% margin) - the number \$6 signs off on:

classical (a CI or a p-value)	not defined
Bayesian posterior (\$4)	0.89 -> TEST FURTHER
re-priced at the placebo floor	0.78 -> TEST FURTHER
re-priced at the raw placebo sd	0.71 -> TEST FURTHER
re-priced at the anchor's SE	0.56 -> TEST FURTHER

('not defined' is no dodge: a p-value ranks data against a null; it attaches no probability to a hypothesis about the effect, and it has never met the \$50k of spend or the 35% margin.)

-> same centre, honest scale: P(profitable) falls from 0.89. The CALL survives (the 0.90 bar is not crossed either way) - but the confidence attached to it was borrowed, not earned.

1 · On the number itself, all three agree — and the model is not the hero. The randomized anchor, the classical SLSQP fit and the Bayesian posterior mean land on top of each other (printed above: the two model-based estimates sit a small fraction of one anchor SE from the referee, well inside its interval). Say this plainly, because it is the whole point of running a companion notebook on real data: **the Bayesian layer did not buy a better point estimate.** It could not have. The causal work was done by the *design* — a coin flip across 50 matched pairs

— and, for the SC arm, by §3’s identification and the simplex. Both estimators share that work; the sampler adds none of it. What the SC arm *does* buy over the anchor is **precision**: the same answer with a far narrower interval, which is exactly why we bother fitting a model on top of a perfectly good experiment — and exactly why we must now ask whether that narrowness is real.

2 · On the uncertainty, they disagree — and the posterior is the optimistic one. This is the uncomfortable finding, and it is measured, not imported. Three numbers price the same object. The posterior’s sd comes from a likelihood that treats the daily gaps as **iid**. The placebo sd comes from re-running *our own estimator* on 25-geo bundles that saw no advertising — the design’s own answer to “how much does this estimator wander when the true effect is zero?” — and even after deflating it for the fact that a 25-geo bundle is noisier than the real 50-geo cell (the conservative floor printed above), it is **~1.6–2.2× wider than the posterior believes**. That is the same failure 07 diagnosed and *quantified* on simulated worlds, where fresh panels were free: there, the posterior **under-priced its own sd by ~2.8×** — and the 90% band it built from that sd went on to cover the planted total only 50% of the time, not 90% (a width error and the coverage shortfall it causes: two statements, not one) — because the synthetic’s error is not fresh noise each day but a **persistent offset** — the unmatched share of the shared factors — so the variance of a *sum* over the window picks up covariance terms the iid likelihood sets to zero. We cannot re-run the world here to measure coverage. We did not need to: the design handed us a scale, and it says the band is too tight. The notebook’s §7 caveat about “interval optimism” was, until this section, an *imported* claim from 07; it is now a **measured** one on this dataset.

One thing this does *not* say, and the print-out is careful about it: the anchor’s SE is **~8×** the posterior’s, and that ratio is **not** the indictment. An unweighted matched-pairs DiD is a genuinely noisier estimator than a donor-matched synthetic control — cutting that variance is the entire job of the model, and it really does cut it. The fair referee for the posterior’s *width* is the placebo distribution of the **same** estimator, and that referee returns a verdict of roughly two, not eight. Both statements are true at once: **synthetic control is much more precise than the raw experiment, and the posterior is still overconfident about how precise.**

3 · What Bayes did buy — and the tension it arrives with. Look at the last block. $P(\text{profitable})$ exists in exactly one column. The classical rows read *not defined*, and that is literal: the DiD’s confidence interval is a statement about a procedure, the permutation p ranks data against a null, and neither has ever met the \$50k of spend or the 35% margin. But §6’s rule is nothing *but* that probability — the margin trap ($\text{iROAS} \times \text{margin} > 1$) has to be integrated over the **whole** distribution of the lift, not evaluated at its endpoints — so the decision quantity lives only in the Bayesian arm. That is real, and it is durable.

And then the honest part: **the decision quantity lives in the arm whose interval we just showed to be too tight.** Re-centre nothing; merely re-price the posterior at the design-based scale, and $P(\text{profitable})$ falls (printed above). In *this* run the verdict survives — TEST FURTHER at every scale, because at a 35% margin we are sitting almost exactly on the break-even iROAS of $1/0.35 = 2.86$ and no amount of re-scaling rescues a knife-edge — but that is luck, not robustness. A run sitting just above the 0.90 GO bar would have had its rollout decision **overturned** by the same correction, and nothing in the posterior’s own output would have warned you.

The ledger, honestly. Bayes did not buy the point estimate (the design already had it, for free, unbiased). It did not buy a trustworthy interval width (it bought a *worse* one than the placebos). It bought the **decision object** — $P(\text{lift} \times \text{margin} > \text{cost})$, propagated end-to-end into dollars, which is the one thing the business actually asked for and the one thing the classical arm cannot

manufacture at any level of effort. The fix for the tension is not “abandon Bayes”; it is what 07 spelled out — *stop feeding a window total to an iid likelihood*: either price the total with the placebo distribution (as this section just did) or give the model correlated errors so it can see the persistence itself. Use the posterior for the decision; use the placebos for the confidence in it. And in the memo, lead with the anchor — the sentence “a randomized experiment says \$147k, and our model agrees” outranks any credible interval in the room.

6 · Decide, in dollars

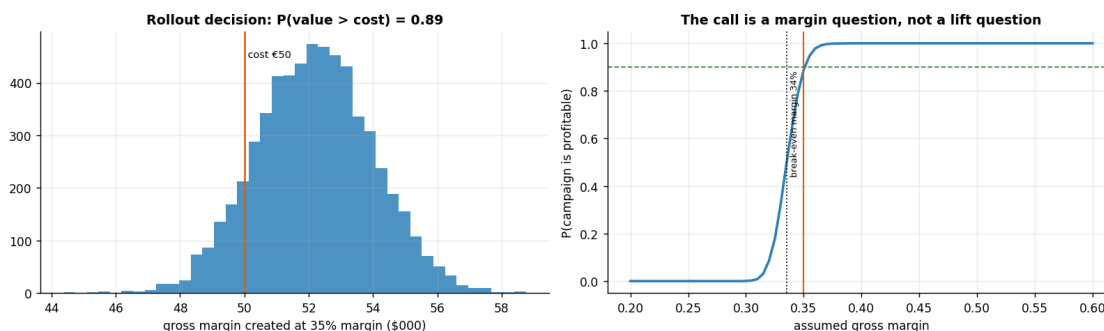
The trap in every iROAS deck: **3.0 revenue-iROAS is not “3× return”**. Ad spend buys *revenue*; the firm keeps only the **gross margin** on it. The campaign pays if

$$\text{iROAS} \times \text{margin} > 1 \quad \Longleftrightarrow \quad \text{iROAS} > \frac{1}{\text{margin}},$$

so at our assumed 35% margin the bar is not 1.0 but **2.86** — and our posterior sits almost on top of it. That is the honest shape of this decision: the *lift* is beyond doubt (Depth B), the *profitability* is a knife-edge that turns entirely on a finance input the model cannot see. The right briefing separates the two.

```
{
  "P_value_gt_cost": 0.889,
  "expected_value": 52.189,
  "expected_net_value": 2.189,
  "expected_roi": 0.044,
  "value_lo": 49.251,
  "value_hi": 55.165,
  "cost": 50.0,
  "decision": "TEST FURTHER"
}
```

break-even margin = $1/\text{iROAS} = 33.5\%$; the campaign clears $P(\text{profitable}) \geq 0.90$ only for margins $\geq 35.5\%$



7 · Caveats

- **The estimate travels; the decision might not.** iROAS ≈ 3 is for *this* creative, *this* spend level, *these* 50 geos, in Feb–Mar. A national rollout extrapolates on all four axes —

diminishing returns alone (nb06’s saturation curves) argue the marginal iROAS at $10\times$ spend is lower.

- **Profitability hinges on an input outside the data.** The margin sweep is the honest deliverable; a single “GO” at 35% margin overstates what the sales panel can know.
- **Spillover is the surviving assumption.** Randomization can’t stop Denver’s ads reaching Boulder’s shoppers; if spend leaks into control geos, both the anchor and SC are biased *downward* (control geos absorb some lift), making our estimate mildly conservative.
- **Interval optimism — measured here, not imported.** 07’s multi-seed check showed SC’s total-lift band under-covers even when the point is right; §5x confirms it on *this* dataset by pricing the posterior’s sd against the placebo distribution of the same estimator (the ratio is printed there). The agreement with the anchor’s *point* is earned; the band’s width is not — quote the placebo interval when the width matters, and read P(profitable) as an upper bound on your confidence.
- **75 interpolated geo-days** (0.8%, small geos, mostly pre-period) — not asserted away: the interpolation check in Step 5 reruns the fit without the gappy donors (a sub-1% shift, printed) and again with every gappy geo dropped outright on both sides — the per-geo lift moves, but the matched-pairs anchor restricted to the same surviving pairs moves with it and the *relative* lift is unchanged: a composition effect, not an interpolation artifact.
- **One experiment.** This is a single draw of the world, and we’ve used its randomization once; the multi-seed calibration story lives in 07 where worlds are free.

Depth A · The naive estimators, on a single real market

In Step 5 the naive comparisons would have looked *fine* — with 50 randomized geos aggregated into a cell, treated-minus-control is unbiased by design; that is precisely why companies run geo experiments. The moment the question becomes 07’s — **one** treated market, no coin flip — the shortcuts break on real data exactly as they broke in simulation. We replay the bake-off pretending only **geo 7** (a mid-size market, ~\$1.8k/day before launch — the cell prints it) ran the campaign, grading against the benchmark the full experiment implies for it (its share of spend $\times$ the anchor iROAS $\Rightarrow \approx +\$394/\text{day}$):

- **before/after** — books the season: sales everywhere rose into spring;
- **treated — average control** — books the *level* gap: geo 7 is nearly $4\times$ **the average control geo’s size** (both measured over the pre-period, both printed below — mixing a pre-period level with a full-period one is how this ratio gets misquoted);
- **DiD vs average control** — fixes the level, keeps the wrong *sensitivity*: a big geo swings more per unit of common shock than the average-of-50, so shared waves don’t cancel;
- **synthetic control** — reweights donors to match geo 7’s own scale and rhythm.

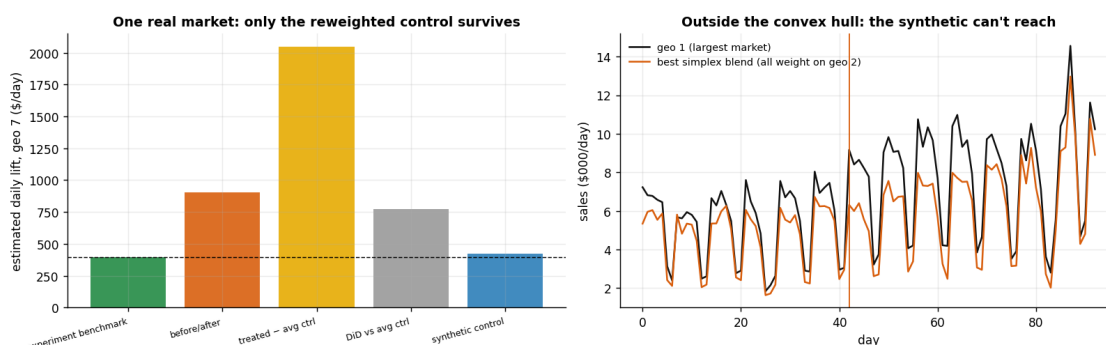
Then the failure 07 could only warn about: **geo 1 is the largest market in the country** — bigger than *every* control geo, hence **outside the convex hull**. The simplex does its honest best (all weight on geo 2, the largest donor), the pre-fit is visibly bad, and the **pre-fit gate FAILS** — the method itself tells you not to trust its number. That self-diagnosis is worth more than the estimate.

geo 7 is a mid-size market: \$1,759/day pre-launch vs \$480/day for the average control geo - 3.7x bigger (both pre-period)

```
{
  "experiment benchmark": 394,
  "before/after": 904,
  "treated \u2212 avg ctrl": 2049,
  "DiD vs avg ctrl": 771,
  "synthetic control": 423
}
```

geo 7 gate: pre-RMSE \$122 vs $|daily\ lift|/3 = \$141 \rightarrow$ PASS (weights spread over ~5 donors)

geo 1 gate: pre-RMSE \$868 vs $|daily\ lift|/3 = \$665 \rightarrow$ FAIL (effective donors 1.0: all weight on geo 2; implied iROAS 5.1 vs the experiment's 2.94 - inflated ~75%)



Read-out. Against geo 7's $\approx +\$394/\text{day}$ benchmark: **before/after** lands at **+\$904** ($2.3\times$ too big — it credits the campaign with spring), **treated — average control** at **+\$2,049** ($5\times$ — it books geo 7's size as lift), **DiD** at **+\$771** ($2\times$ — level fixed, sensitivity still wrong: a \$1.8k/day market swings harder per common shock than the \$0.5k/day average control does — the printed pre-period ratio is $\sim 3.7\times$), and **synthetic control** at **+\$423**, **within ~7%**, with the gate PASSing (pre-RMSE \$122 vs the \$141 bar). Same data, four estimates, one survivor — 07's bake-off, no simulator required. And geo 1 is the counterexample that keeps us honest: the country's largest market has no counterfactual among smaller donors, the gate FAILs loudly (pre-RMSE \$868 vs a \$665 bar), and the estimate it would have produced — iROAS ≈ 5.1 vs the experiment's 2.94 — is exactly the number an unscrupulous deck would have shipped. **When the treated unit escapes the donor hull, aggregate or walk away.**

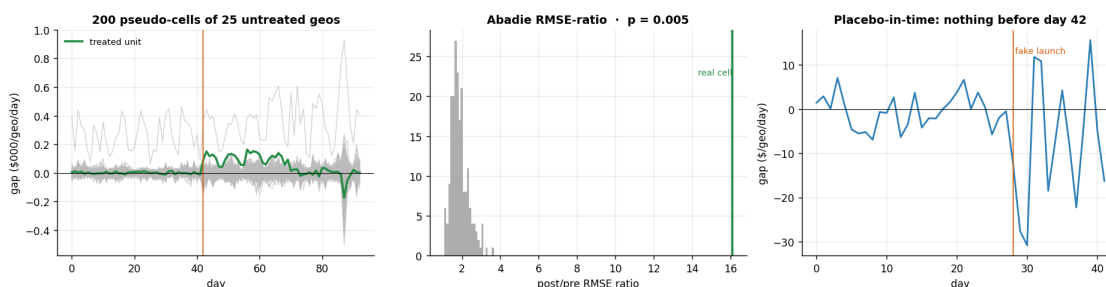
Depth B · Placebo inference — is the gap statistically real?

Step 0 already ran this test and printed its headline p-values; here we open the box. The logic is 07's, adapted to the design. The treated unit is a *cell* — the mean of 50 geos — so the fair placebo is not a single geo but a **pseudo-cell**: draw 25 of the 50 *control* geos at random (the largest cell that still leaves a donor pool), pretend they were a campaign cell, fit them a synthetic control from the remaining 25, and record the post-window “gap.” None of these bundles saw a single ad dollar, so their gaps map out pure noise — **the permutation distribution the design itself**

generates. One scale mismatch to own: a mean of 25 geos is noisier than the real cell’s mean of 50 (roughly $2\times$ the idiosyncratic variance), and each placebo fit gets only 25 donors instead of 50 — both push the placebo gaps *wider* than the real cell’s own null, so the permutation p is, if anything, **conservative**; a real cell that still tops these noisier placebos has cleared a higher bar than the nominal p suggests. Two test statistics, as in 07: the raw mean post-gap, and Abadie’s scale-free **post/pre RMSE ratio** (which doesn’t reward bundles that simply fit badly everywhere). We also rerun 07’s **placebo-in-time** — a fake launch 14 days before the real one, inside the pre-period.

This permutation test is the *calibrated* significance statement for this notebook — the posterior band is the optimistic one (07’s lesson, and §5x measured it here), the rank among placebos is honest by construction. The same 200 pseudo-cells also gave Step 0 its randomization interval and §5x its design-based scale: one null, three jobs.

pseudo-cell permutation: real gap \$+105/geo/day vs placebo 95% range [-17, +21]
 -> p = 0.010
 RMSE-ratio test: real cell 16.1 vs placebo median 1.78 -> p = 0.005 (no placebo bundle comes close: the real cell ranks 1st of all 201 units)
 placebo-in-time: fake-launch gap \$-8/geo/day - 7% of the real \$+107 effect and the wrong sign (pre-fit RMSE \$4) - no anticipation, no invented lift



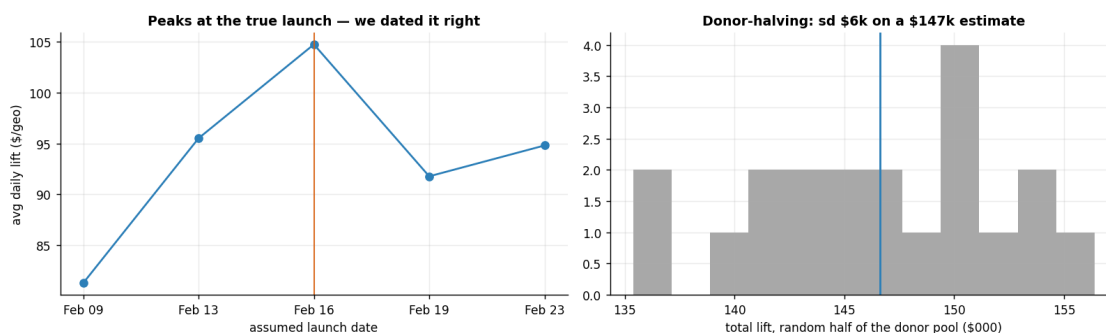
Read-out. The spaghetti plot says it before any statistic does: the real cell’s post-launch gap ($\approx +\$105/\text{geo}/\text{day}$, printed above) sits far outside anything the untreated pseudo-cells produce, and Abadie’s scale-free RMSE ratio — the real cell near 16 against a placebo median near 2 — removes the objection that it merely fits badly everywhere. Read the printed p-values carefully, though. First, a permutation p has a *floor* of $1/(N_{\text{pseudo}} + 1)$ — $1/201 \approx 0.005$ at the FULL run’s 200 placebos — so hitting the floor means “the most extreme unit we observed,” not an arbitrarily small tail probability; the print above reports the real cell’s actual rank rather than asserting it. Second, recall the scale mismatch flagged above: these 25-geo placebos are *noisier* than the 50-geo real cell, so the printed p is conservative. The placebo-in-time closes the loop: a fake launch 14 days inside the pre-period finds $\approx -\$8/\text{geo}/\text{day}$ — under a tenth of the real effect, and the wrong sign — no anticipation, no invented lift.

Depth C · The dollar decision & sensitivity

Three stress tests on the number a CMO would repeat in a board meeting, then the verdict.
Launch-date sweep: as in 07, mis-dating the campaign folds no-effect days into the window

and dilutes the average — the estimate should peak *at* the true launch, and the peak is itself a diagnostic that we dated it right. **Donor-pool halving**: refit on random halves of the donor pool; a result that owns its donor pool shouldn't lurch. **The margin $\times$ iROAS surface** is already in Step 6; here we print the summary a decision memo needs.

```
{
  "total_incremental_sales_k": 149.1,
  "ci90_k": [
    140.7,
    157.6
  ],
  "iroas": 2.98,
  "did_anchor_k": 147.2,
  "did_anchor_se_k": 41.9,
  "pre_rmse_gate": "PASS",
  "p_pseudo_gap": 0.01,
  "p_rmse_ratio": 0.005,
  "placebo_in_time_gap": -7.8,
  "cooldown_gap_per_day": -2.5,
  "break_even_margin": 0.335,
  "P_profitable_at_35pct_margin": 0.889,
  "decision_at_35pct": "TEST FURTHER"
}
```



The one-paragraph decision

The campaign **worked**, and we can prove it to **experimental standard**: the synthetic control tracks the treated cell to within $\sim \$10/\text{geo}/\text{day}$ before launch (gate PASS), lands on the randomized matched-pairs answer ($\$149\text{k}$ vs $\$147\text{k}$ of incremental sales; iROAS 2.98 vs 2.94), is the most extreme unit on both placebo permutation tests ($p \approx 0.005\text{--}0.010$), shows nothing material at a fake launch date ($-\$8/\text{geo}/\text{day}$, 7% of the real effect, wrong sign), survives leave-one-out and donor-halving, and — the real-data bonus no simulator provides — **switches off within days of the spend stopping**, which both certifies the attribution and denies any carryover annuity. What the data *cannot* settle is profitability: $\$2.98$ of revenue per ad dollar clears cost only above a **33.5% gross margin**, and reaches 90% confidence only above $\sim 35.5\%$. So the

briefing is two sentences, not one: *the lift is real and precisely measured; whether to scale is a margin question* — GO for the high-margin lines, NO-GO below one-in-three margins, and any national rollout should expect the marginal iROAS to sag below this test's 2.98 (saturation, notebook 06) — so re-test at scale rather than extrapolate.

Book export

The book chapter built from this notebook (`book/chapters/geo_lift.tex`) does not retype a single number. Every figure it prints — every euro amount, coverage rate, p-value, table cell and plot — is *emitted* below by `cmp.report` into `book/build/results.json`, from which `book/build.py` generates `macros.tex`, the booktabs tables and the vector figures the chapter `\inputs`. A number that goes stale is impossible; a number that goes *missing* is a LaTeX compile error rather than a silent hole. The cell adds no output of its own — the notebook above is unchanged.

book export: 101 keys emitted to `book/build/results.json`